

SIMATS ENGINEERING

SWETHA.B

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EXPERIMENT 18

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MACHINE LEARNING

Implement Perceptron based IRIS classification

Code

```
# Perceptron based Iris Classification

from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import Perceptron
from sklearn.metrics import accuracy_score, confusion_matrix, ConfusionMatrixDisplay
import matplotlib.pyplot as plt

# Load Iris dataset
iris = load_iris()

X = iris.data
y = iris.target

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Scale features
scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

# Create Perceptron model
model = Perceptron(
    max_iter=1000,
    random_state=42
)

# Train the model
model.fit(X_train, y_train)
```

```
# Predict
y_pred = model.predict(X_test)

# Calculate accuracy
accuracy = accuracy_score(y_test, y_pred)

print("Accuracy:", accuracy)
print("Accuracy (%):", accuracy * 100)

# Confusion Matrix
cm = confusion_matrix(y_test, y_pred)

print("\nConfusion Matrix:")
print(cm)

ConfusionMatrixDisplay(
    confusion_matrix=cm,
    display_labels=iris.target_names
).plot()

plt.title("Perceptron Based Iris Classification")
plt.show()
```

OUTPUT:

```
*** Accuracy: 0.9333333333333333
    Accuracy (%): 93.33333333333333
```

Confusion Matrix:

```
[[10  0  0]
 [ 0  9  0]
 [ 0  2  9]]
```

